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Chance and the Arrow

DOES TIME HAVE A DIRECTION?

Even if time doesn't flow, it still makes sense to ask whether it has an arrow—whether there is a direction to the way things unfold in time that can be discerned in the laws of physics. It is the question of whether there is some intrinsic order in how events are sprinkled along spacetime and whether there is an essential scientific difference between one ordering of events and the reverse ordering. As everyone already knows, there certainly appears to be a huge distinction of this sort; it's what gives life promise and makes experience poignant. Yet, as we'll see, explaining the distinction between past and future is harder than you'd think. Rather remarkably, the answer we'll settle upon is intimately bound up with the precise conditions at the origin of the universe.

The Puzzle

A thousand times a day, our experiences reveal a distinction between things unfolding one way in time and the reverse. A piping hot pizza cools down en route from Domino's, but we never find a pizza arriving hotter than when it was removed from the oven. Cream stirred into coffee forms a uniformly tan liquid, but we never see a cup of light coffee unstir and separate into white cream and black coffee. Eggs fall, cracking and splattering, but we never see splattered eggs and eggshells gather together and coalesce into uncracked eggs. The compressed carbon dioxide gas in a bottle of Coke rushes outward when we twist off the cap, but we never find spread-out carbon dioxide gas gathering together and swooshing back into the bottle. Ice cubes put into a glass of room-temperature water melt, but we never see globules in a room-temperature glass of water coalesce into solid cubes of ice. These common sequences of events, as well as countless others, happen in only